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THE BRADLEY–TERRY STOCHASTIC BLOCK MODEL

BY LAPO SANTI[1,a], NIAL FRIEL[2,b]

[1] *School of Mathematics and Statistics, University College Dublin, Dublin, Ireland,* [a] *lapo.santi@ucdconnect.ie*

[2] *Insight Centre for Data Analytics, University College Dublin, Dublin, Ireland,* [b] *nial.friel@ucd.ie*

The Bradley–Terry model is widely used for the analysis of pairwise comparison data and, in essence, produces a ranking of the items under comparison. We embed the Bradley–Terry model within a stochastic block model, allowing items to cluster. The resulting Bradley–Terry SBM (BT–SBM) ranks clusters so that items within a cluster share the same tied rank. We develop a fully Bayesian specification in which all quantities—the number of blocks, their strengths, and item assignments—are jointly learned via a fast Gibbs sampler derived through a Thurstonian data augmentation. Despite its efficiency, the sampler yields coherent and interpretable posterior summaries for all model components. Our motivating application analyses men's tennis results from ATP tournaments from the 2000 season up to the 2025 season. We find that the top 105 players can be broadly partitioned into three or four tiers in most seasons. Moreover, the size of the strongest tier was small from the mid-2000s to 2018. Between 2019 and 2022, we observe a transition period characterised by a gradual widening of the top tier, while in more recent seasons (2023–2025) the structure appears to revert to a more elite configuration, coinciding with the rise of dominant players such as Carlos Alcaraz and Jannik Sinner.

1. Introduction. Rankings are ubiquitous. From institutional league tables in education and health to online reviews, sports tournaments and app ratings, ordered lists routinely inform decisions, allocate resources, and shape reputations. Their appeal lies in their simplicity: complex information about each item is distilled into a single number. Yet this simplicity can be misleading, creating a false sense of precision and granularity that the data often do not support. As Goldstein and Spiegelhalter (1996) famously cautioned, there are *"quantifiable uncertainties which place inherent limitations on the precision with which institutions can be compared"*, a critique that extends beyond schools and hospitals to rankings of all kinds.

These tensions are particularly evident in competitive sports, where rankings are both economically consequential and statistically fragile. In professional tennis – the substantive focus of this paper – rankings are central. A seemingly minor drop in position can alter tournament seedings, jeopardise endorsement deals, affect a player's self-image, and ultimately determine whether a career remains sustainable (Schöttl et al., 2025). Rankings depend on players' performances across tournaments, but it remains unclear to what extent a drop in rank reflects a genuine performance gap.

Ranking lists in many contexts, particularly sports, are typically strict, in the sense that if two players are selected at random, one player will always be ranked higher than the other. In so doing, such a ranking produces a strict complete ordering of the items (for example, players) under consideration and implies that tied ranks are not permitted. In professional tennis, this assumption can be particularly brittle. Two players separated by a single rank may never have competed head-to-head, may have followed markedly different tournament schedules, or may have been subject to different quality of opponents. A strict complete ordering nonetheless

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treats these noisy and context-dependent differences as meaningful, potentially overstating separation where none is credibly supported by the data.

The limitations of strict rankings have only recently begun to be addressed in Bayesian approaches (Pearce and Erosheva, 2025; Piancastelli and Friel, 2024), which go beyond strict total orders by accommodating tied ranks of items. Building on this line of work, our objective is to develop a model that clusters items that share the same rank while imposing a strict ordering only on the cluster labels. This yields a coarser ranking, but it is often a more faithful representation of uncertainty: rather than forcing spurious pairwise distinctions, players of similar abilities may be assigned to the same cluster.

To operationalise this idea, we take the Bradley–Terry (BT) model (Bradley and Terry, 1952) as our starting point. The BT framework assigns each player a latent strength (or ability) parameter. Sorting these parameters yields a ranking of players, thereby inducing a strict total order. In other words, the BT model inherits the very limitation we aim to overcome: it treats even negligible differences in estimated strengths as meaningful, producing a strictly ordered list of players.

To relax this assumption, we extend the BT framework by embedding it within the Stochastic Block Model (SBM) (Holland, Laskey and Leinhardt, 1983; Nowicki and Snijders, 2001). This combination preserves the well-known BT likelihood while introducing a latent block structure that clusters players into groups of similar strength. The resulting model replaces spurious distinctions with a clustered ordering in which only the clusters themselves are strictly ranked. This formulation, which combines pairwise comparison modelling with latent clustering, directly motivates the following methodological contributions.

We provide an *end-to-end probabilistic analysis of pairwise-data networks:* we model pairwise outcomes as a directed comparison network and make *ranking itself a random object*. The BT likelihood is retained, but ranking is transferred from individual items to *ordered blocks*. The result is a fully generative Bayesian model in which posterior samples propagate uncertainty through every layer, from block membership to block strengths and even the number of blocks, so that summaries and decisions are *probabilistic by design* rather than point estimates.

We allow for a *finite yet data-driven number of blocks:* a nonparametric Gibbs-type (Gnedin) prior partitions items into a random but finite number of blocks, with reinforcement dynamics well suited to bounded, tournament-like populations such as seasonal sports leagues. This prior combines parsimony with heavy-tailed flexibility and lets K, the number of clusters, be learned from the data.

We develop *fully conjugate inference with variable dimension:* a Thurstonian data augmentation (Caron and Doucet, 2012) yields closed-form full conditionals and a fast single-site Gibbs sampler that *jointly* learns the number of blocks, their strengths, and player memberships, traversing a variable-dimensional posterior *without* reversible-jump or split–merge moves, with per-iteration cost $\mathcal{O}(|E| + nK)$, where E is the total number of observed dyads.

We obtain *uncertainty-aware posterior summaries:* from the same posterior draws, we derive fully probabilistic and internally coherent quantities that directly address applied questions, the probability that a player belongs to a given tier, the distribution over K, odds-interpretable block strengths, and entropy-based measures of competitive balance with credible intervals. These posterior quantities are designed for reporting, comparison, and principled decision-making.

Empirically, we apply the BT–SBM to twenty-six seasons (2000 to 2025, both included) of the results of matches from the *Association of Tennis Professionals* (ATP), focusing on the top $n = 105$ players each year. The posterior summaries reveal a clear and mutually reinforcing narrative: during the so-called "Big Four" era (when the players Federer, Nadal, Djokovic, and Murray dominated), roughly the 2006–2018 seasons, the elite field contracts and stabilises,

producing a sharp concentration of strength, reflected in a small number of players assigned to the strongest cluster. Between 2019 and 2022, we observe a transition phase in which the distribution of posterior cluster memberships broadens, indicating a more open and less hierarchically structured competitive landscape. In the most recent seasons (2023–2025), however, the elite tier appears to contract again, consistent with the emergence of a new generation of dominant players, most notably Carlos Alcaraz and Jannik Sinner. Beyond interpretability, the BT–SBM also improves predictive accuracy relative to the standard Bradley–Terry model, confirming the practical value of block-ranking as a robust, data-supported alternative to noisy individual rankings.

Taken together, our results make the case for moving beyond strict complete orderings toward clustered, uncertainty-aware rankings that offer a more interpretable and empirically faithful view of pairwise data.

2. Motivating application and dataset: men's ATP results data. The substantive application which motivates our work concerns men's professional tennis. Specifically, we consider results of matches from the *Association of Tennis Professionals* (ATP) men's tour[1]. The ATP men's tour comprises a series of tournaments. A feature of the ATP series is that the performance of each player in a tournament is then used to determine an ATP rank of each player. As highlighted above, this is a strict complete ranking. We focus on an analysis of matches over 26 seasons from 2000 through to 2025, where we use the 2017 season as an illustrative example. Throughout this study period, we fixed the number of players each season to $n = 105$, to ensure comparability across seasons.

For each season, we record the outcome of each match as an ordered pair between two players. Let w_{ij} denote the number of times that player i defeated player j in that season, and let $n_{ij} = w_{ij} + w_{ji}$ be the total number of matches played between players i and j. Since most pairs never meet in a given season, we explicitly encode partial observation through a binary mask $M_{ij} = \mathbb{I}\{n_{ij} > 0\}$, for $i \neq j$, and define the set of observed dyads as $E = \{(i, j) : i < j,\ M_{ij} = 1\}$.

We collect the win counts in a matrix $\mathbf{W}$ of size $n \times n$, and the match-count matrix $\mathbf{N} = \mathbf{W} + \mathbf{W}^\top$, so that $\mathbf{N}$ is symmetric and $n_{ii} = 0$ for all i as a player cannot play against themselves. Dyads with $M_{ij} = 0$ are treated as structurally unobserved rather than missing outcomes.

Figure 1 presents a graphical representation of $\mathbf{W}$, with players ordered along both axes according to their ATP ranking at the end of the season 2017. Each cell encodes the number of wins of the row player against the column opponent, using a colour gradient. White pixels indicate unobserved pairs, i.e., players that did not play against each other in that season, and the dark-to-light discrete palette represents increasing win counts. On the right, we show the total number of matches played by each player.

Several structural features emerge from an inspection of Figure 1. The matrix is sparse – most pairs of players never meet – and this sparsity reflects the well-known "winners-play-more" effect of knockout tournaments: lower-ranked players (in the bottom-right corner) face each other less often, while top players (top-left) meet frequently. The matrix is also skew-symmetric along the main diagonal. In the top-right corner, matches predominantly result in wins for higher-ranked players; in the bottom-left, we mainly observe losses. Along the diagonal band, by contrast, outcomes tend to be more balanced, with reciprocal results and local cycles that are inconsistent with the notion of a strict global ordering.

Just as important is what the pairwise comparison matrix fails to capture. Each cell simply counts wins but ignores contextual factors: the timing of the match (early vs. late season), the

[1] See https://www.atptour.com/ for further details.

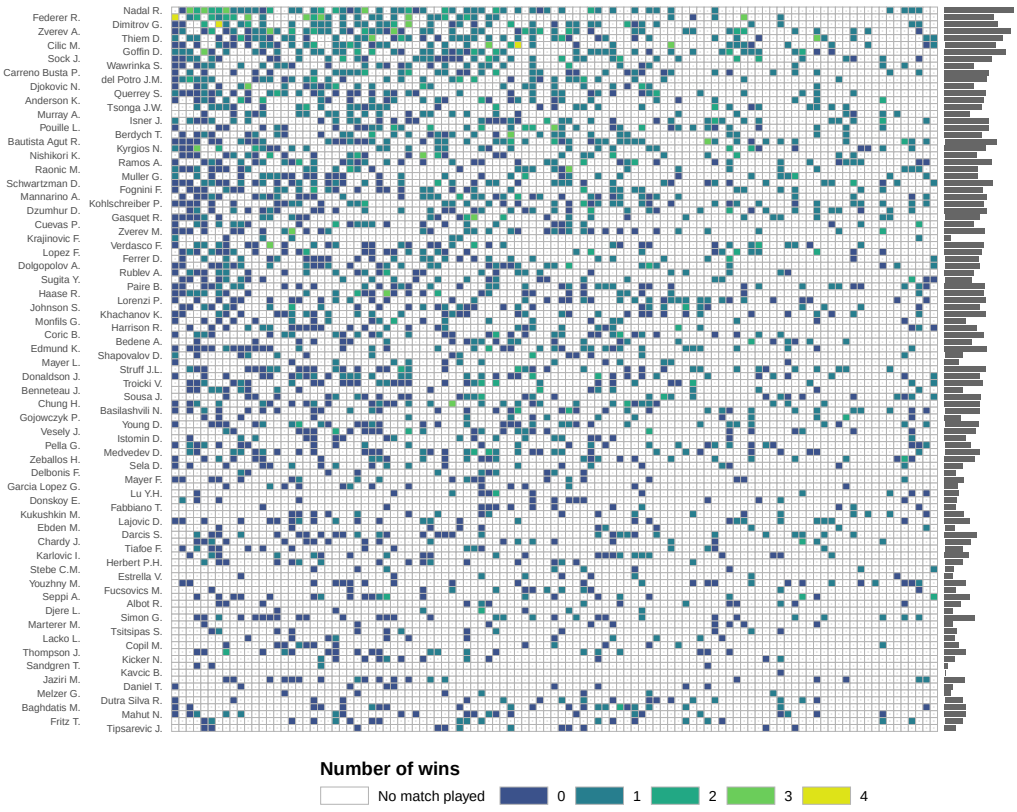


Fig 1: The match outcome matrix $\mathbf{W}$ for the ATP 2017 season represented as an adjacency matrix where the players are ordered by their end-of-season ATP ranking. Each cell represents the number of wins of the row player against the column player. Number of wins are displayed using a dark to light discrete gradient; white cells denote unobserved pairs of players. The main diagonal is unobserved, as self-edges are not possible in the current setting. Right margin bars show total matches played. The names on the y-axis are staggered across two columns to improve readability.

surface (clay, grass, hard), the physical condition of the player (injuries, fatigue), or situational aspects (home advantage, retirements). These structural features – sparsity, skew-symmetry, and local inconsistencies – motivate moving beyond a strict one-dimensional ordering to a model that explicitly accommodates uncertainty, allowing for tied or clustered ranks when the data do not credibly support fine distinctions.

3. Literature review. We briefly review the contributions closest to ours. At the core lies the Bradley–Terry (BT) model, which assigns the probability that item i defeats item j as

$$p(i \text{ beats } j \mid \boldsymbol{\lambda}) = \frac{\lambda_i}{\lambda_i + \lambda_j}, \qquad \lambda_i, \lambda_j > 0 \quad \text{for all } (i,j) \in E, \tag{1}$$

where $\boldsymbol{\lambda} = \{\lambda_1, \ldots, \lambda_n\}$ is the vector of item-specific strengths. The BT model remains a cornerstone of paired-comparison modelling and has generated a large literature (Croon and Luijkx, 1993; Glickman, 2008; Caron and Doucet, 2012; Whelan, 2017; Seymour et al., 2020; Wainer, 2023).

Ordering the parameters of $\boldsymbol{\lambda}$ yields an implicit ranking: if $\lambda_i > \lambda_j$, then the probability that i defeats j exceeds 0.5, and sorting the strengths produces a complete linear order of the

items. The classical BT model therefore enforces a strict ranking through the latent strengths. Such order is characterised by the property of linear stochastic transitivity (LST): if $\lambda_i > \lambda_j$ and $\lambda_j > \lambda_k$, then necessarily $\lambda_i > \lambda_k$, so all items can be placed on a single latent strength scale.

In tennis, outcomes also depend on surface, injuries, schedule, and local form. Classical BT extensions absorb part of this heterogeneity through covariates in the log-odds of victory (Agresti, 2013; Turner and Firth, 2012; Cattelan, 2012). We take a different route: we introduce a latent partition $\mathbf{x}$ so that players in the same cluster share the same strength. This lets equally-strong players remain tied, while the ordering shifts at the cluster level.

Several authors have brought clustering into paired-comparison models. On the frequentist side, Masarotto and Varin (2012) use a ranking lasso to fuse similar strengths, and Tutz and Schauberger (2015) study ordered paired-comparison models with structured grouping effects. On the Bayesian side, Wu et al. (2015) use mixtures to capture heterogeneous preference patterns, Hsiao and Wang (2021) propose a "Cluster-BT" model where within-cluster comparisons follow BT and between-cluster comparisons are treated as noise, and Spearing et al. (2021) cluster both items and interactions to explain inconsistent outcomes. Our focus is narrower: we seek item-level clusters that can be read as tied tiers in a transitive structure.

The closest comparison is Pearce and Erosheva (2025), who also infer latent blocks with shared strengths in a fully generative Bayesian BT model, using a Partition-based Spike-and-Slab Fusion prior. The main difference sits in the partition prior. Their prior depends only on the number of clusters, so partitions with the same K_g receive the same prior mass. We instead use a Gnedin prior, which also carries information on cluster sizes through an urn scheme. This buys us simple Gibbs updates, inference on K without reversible-jump proposals, and considerably faster computation. We compare the two approaches in the simulation study.

4. Embedding the BT model within an SBM. Competitions are inherently relational systems, and network-based representations have long been used to analyse competitive settings. In team sports, for example, such approaches are employed to assess relative contributions within the same team, for instance through plus-minus-based network models (Josephs and Upton, 2025; Piette, Pham and Anand, 2011). Our setting differs in that we focus on paired comparison data among rival tennis players that are represented as a weighted directed network where edges encode wins and losses.

Stochastic Block Models (SBMs) provide a probabilistic framework to uncover clusters of nodes that exhibit similar connection probabilities. They have been used to study sports settings before, as football leagues (Basini et al., 2023), chess tournaments and American football (Vaca-Ramírez and Peixoto, 2022), among others.

Building on these ideas, we introduce the Bradley–Terry Stochastic Block Model (BT–SBM), which treats tiers of players of equal strength as blocks in a SBM, while retaining the probabilistic comparison framework of Bradley–Terry. We briefly review the main SBM features before detailing the full model.

4.1. *Overview of SBMs.* Let $\mathbf{x} = (x_1, \ldots, x_n)$ denote latent block assignments with $x_i \in \{1, \ldots, K\}$. Nodes in the same block have equal connection probability with other blocks, a distinctive feature of SBMs called *stochastic equivalence*. In the canonical SBM (Nowicki and Snijders, 2001), the edge presence is modelled as a binary indicator

$$y_{ij} \mid \mathbf{x}, \boldsymbol{\theta} \sim \text{Bernoulli}(\theta_{x_i x_j}),$$

where $\boldsymbol{\theta} = (\theta_{rs})_{r,s=1}^{K}$ is the block-connectivity matrix. Each θ_{rs} is typically assigned an independent $\text{Beta}(a, b)$ prior. The allocation vector $\mathbf{x}$ is often modelled via mixing proportions

$\boldsymbol{\pi} \sim \mathrm{Dirichlet}(\alpha_1, \ldots, \alpha_K)$, inducing a Dirichlet–multinomial prior on $\mathbf{x}$, after marginalising over $\boldsymbol{\pi}$.

Recent developments in Bayesian nonparametrics have introduced Gibbs-type priors for exchangeable random partitions, including the Dirichlet–multinomial model as a special case (Pitman, 1996; De Blasi et al., 2015). Prominent examples are the Dirichlet process, the Pitman–Yor process, and the Gnedin prior. They admit a sequential predictive urn representation and explicit control over the growth of the number of clusters and the distribution of block sizes; see De Blasi et al. (2015) and Legramanti et al. (2022).

4.2. *A Gnedin Prior for the Latent Partition.* We replace the usual Dirichlet–multinomial prior with a Gibbs-type prior,

$$p(\mathbf{x}) = \psi_{n,K(\mathbf{x})} \prod_{k=1}^{K(\mathbf{x})} (1-\sigma)_{m_k - 1}, \quad \sigma < 1, \tag{2}$$

where $m_k = |\{i : x_i = k\}|$, $K(\mathbf{x})$ is the number of blocks, and $(a)_r$ denotes the ascending factorial. We write K for simplicity.

The triangular recursion

$$\psi_{n,K} = (n - \sigma K)\, \psi_{n+1,K} + \psi_{n+1,K+1}, \quad \psi_{1,1} = 1,$$

induces the sequential predictive urn scheme (Lijoi, Mena and Prünster, 2007)

$$p(x_{n+1} = k \mid \mathbf{x}) \propto \begin{cases} \psi_{n+1,K}\,(m_k - \sigma), & k = 1, \ldots, K, \\ \psi_{n+1,K+1}, & k = K+1. \end{cases}$$

Within this class, we adopt the *Gnedin* prior and write $\mathbf{x} \sim p_{\mathrm{GN}}(\mathbf{x} \mid \gamma)$, where p_{GN} denotes the Gibbs-type prior with $\sigma = -1$ and parameter $\gamma \in (0, 1)$. Larger values of γ penalise the creation of additional clusters. In this case,

$$\psi_{n,K} = \frac{(\gamma)_{n-K} \prod_{k=1}^{K-1} (k^2 - \gamma k)}{\prod_{i=1}^{n-1} (i^2 + \gamma i)},$$

and the predictive urn scheme simplifies to

$$p(x_{n+1} = k \mid \mathbf{x}) \propto (m_k + 1)(n - K + \gamma), \quad p(x_{n+1} = K+1 \mid \mathbf{x}) \propto K^2 - K\gamma. \tag{3}$$

This representation buys us simple updates for $(\mathbf{x}, K)$ and competitive empirical behaviour relative to other Gibbs-type priors (Legramanti et al., 2022; Lu, Durante and Friel, 2025). From a modelling perspective, the *population* number of latent blocks is random but almost surely finite, which is natural here because within a season we expect only a limited number of strength tiers. The prior has mode at $k = 1$, heavy tails, and infinite expectation (Legramanti et al., 2022), so it favours parsimonious solutions while still assigning non-negligible mass to larger values of K. Its reinforcement mechanism also favours the growth of already established groups, leading to imbalanced cluster sizes — a small elite and a large lower tier, as in professional tennis.

Another useful property is the *closed analytic form* of the prior distribution of K:

$$p(K = k \mid n, \gamma) = \binom{n}{k} \frac{(1-\gamma)_{k-1} (\gamma)_{n-k}}{(1+\gamma)_{n-1}}, \quad k = 1, \ldots, n.$$

Hence,

$$\mathbb{E}[K \mid n, \gamma] = \frac{\Gamma(n+1)\Gamma(1+\gamma)}{\Gamma(n+\gamma)}, \quad \mathrm{Var}[K \mid n, \gamma] = \mathbb{E}[K]\big[n - \gamma(n-1)\big] - \mathbb{E}[K]^2. \tag{4}$$

This compact expression shows how the variability of the number of clusters depends jointly on n and γ. Lower values of γ produce a heavier-tailed prior on K and therefore make additional clusters more likely, even at moderate sample sizes. Further discussion of (4) is deferred to Appendix C.

4.3. *Block-clustered BT likelihood.* Given $\mathbf{x} \sim p_{\mathrm{GN}}(\mathbf{x} \mid \gamma)$, each block is assigned a strength parameter λ_k, and we collect them as $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_K)$. We assign independent Gamma priors,

$$\lambda_k \mid \mathbf{x} \sim \Gamma(a, b), \tag{5}$$

a conjugate choice in Bradley–Terry models (Gormley and Murphy, 2009-06; Guiver and Snelson, 2009). Conditional on $\mathbf{x}$ and $\boldsymbol{\lambda}$,

$$p(i \text{ beats } j \mid \mathbf{x}, \boldsymbol{\lambda}) = \frac{\lambda_{x_i}}{\lambda_{x_i} + \lambda_{x_j}}.$$

The likelihood is therefore

$$\mathcal{L}(\mathbf{W} \mid \boldsymbol{\lambda}, \mathbf{x}, \mathbf{N}) \propto \prod_{i<j} \left[\left(\frac{\lambda_{x_i}}{\lambda_{x_i} + \lambda_{x_j}} \right)^{w_{ij}} \left(\frac{\lambda_{x_j}}{\lambda_{x_i} + \lambda_{x_j}} \right)^{n_{ij} - w_{ij}} \right]^{M_{ij}}. \tag{6}$$

The observation mask $M_{ij} = 1$ corresponds to $n_{ij} > 0$, and therefore the likelihood only includes observed dyads, as in Shen et al. (2025).

4.4. *Data augmentation and conjugacy.* Sampling from the posterior distribution

$$p(\boldsymbol{\lambda}, \mathbf{x} \mid \mathbf{W}, \mathbf{N}) \propto \underbrace{\mathcal{L}(\mathbf{W} \mid \boldsymbol{\lambda}, \mathbf{x}, \mathbf{N})}_{\text{BT-likelihood as in (6)}} \times \underbrace{p(\boldsymbol{\lambda} \mid \mathbf{x})}_{\text{Strengths' prior as in (5)}} \times \underbrace{p_{\mathrm{GN}}(\mathbf{x} \mid \gamma)}_{\text{Gnedin prior spec. of (2)}}$$

via a standard Metropolis-within-Gibbs is not straightforward. In particular, this leads to poor mixing and consequently slow convergence to the posterior distribution. To address this issue, we follow Caron and Doucet (2012), who reinterpret each individual comparison as a continuous-time event, by introducing the latent variables:

$$Y_{k,i} \sim \mathrm{Exp}(\lambda_{x_i}), \quad Y_{k,j} \sim \mathrm{Exp}(\lambda_{x_j}), \quad k = 1, \ldots, n_{ij},$$

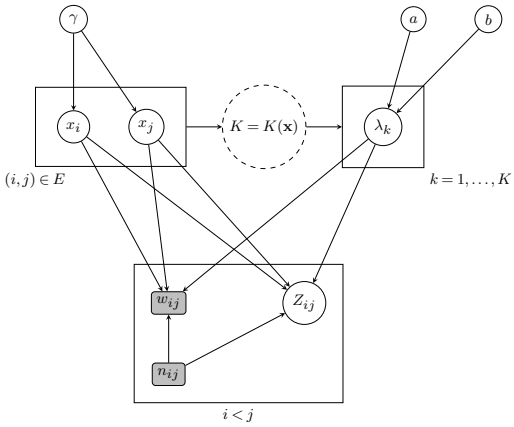
so that $p(i \text{ beats } j) = p(Y_{k,i} < Y_{k,j}) = \frac{\lambda_{x_i}}{\lambda_{x_i} + \lambda_{x_j}}$. Define

$$Z_{ij} = \sum_{k=1}^{n_{ij}} \min\{Y_{k,i}, Y_{k,j}\} \sim \Gamma(n_{ij}, \lambda_{x_i} + \lambda_{x_j}),$$

which yields the augmented likelihood

$$\mathcal{L}(\mathbf{W}, \mathbf{Z} \mid \boldsymbol{\lambda}, \mathbf{x}, \mathbf{N}) \propto \prod_{i<j} \left[\lambda_{x_i}^{w_{ij}} \lambda_{x_j}^{n_{ij} - w_{ij}} \frac{Z_{ij}^{n_{ij}-1}}{\Gamma(n_{ij})} \exp\!\big[-(\lambda_{x_i} + \lambda_{x_j}) Z_{ij}\big] \right]^{M_{ij}}. \tag{7}$$

This restores conjugacy and enables a convenient posterior inference via a fast Gibbs sampler.



Require: n, $\mathbf{N} = \{n_{ij}\}_{i<j}$, a, b, γ.
Initialise: $\{K, x_1\} \leftarrow 1, \lambda_1 \sim \Gamma(a, b)$;
for $i = 2$ **to** n **do**
  $x_i \sim \mathrm{GN}(\gamma)$ as in (3)
  if $x_i = K + 1$ **then**
    $K \leftarrow K + 1$; $\lambda_K \sim \Gamma(a, b)$
  end if
end for
for all pairs $i < j$ with $M_{ij} = 1$ **do**
  $w_{ij} \mid n_{ij}, \mathbf{x}, \boldsymbol{\lambda} \sim \mathrm{Bin}\left(n_{ij}, \frac{\lambda_{x_i}}{\lambda_{x_i} + \lambda_{x_j}}\right)$
end for

Fig 2: On the left, the directed acyclic graph (DAG) highlights the hierarchical structure of the model: hyperparameters (a, b) and γ govern the latent cluster strengths $\boldsymbol{\lambda}$ and the item-specific allocations $\mathbf{x}$, respectively. These in turn determine the distribution of the observed match outcomes $\mathbf{W} = (w_{ij})$ given the number of matches $\mathbf{N} = (n_{ij})$. The number of occupied blocks K is determined by $\mathbf{x}$, and K affects the size of the plate for λ_k. The grey-coloured boxes represent observed outcomes. On the right, the algorithm sketches the generative procedure required to sample from the BT–SBM. Items are allocated one at a time either to an existing block or to a newly created one according to the predictive distribution $p(x_{n+1} \mid \mathbf{x})$ in Eq. (3), while the number of occupied blocks K evolves accordingly. After block assignments and cluster strengths have been sampled, match outcomes are generated conditionally on $(\mathbf{x}, \boldsymbol{\lambda}, \mathbf{N})$. This procedure yields partitions with the characteristic features of the Gnedin prior, namely, a small modal number of clusters and a heavy-tailed distribution, and match outcomes reflecting K different tiers of strength.

4.5. *Full Posterior Distribution.* Combining the above elements, we obtain the full posterior,

$$p(\boldsymbol{\lambda}, \mathbf{x}, \mathbf{Z} \mid \mathbf{W}, \mathbf{N}) \propto \prod_{i<j} \left[\frac{\lambda_{x_i}^{w_{ij}} \lambda_{x_j}^{n_{ij}-w_{ij}}}{\Gamma(n_{ij})} Z_{ij}^{n_{ij}-1} \exp\{-(\lambda_{x_i} + \lambda_{x_j}) Z_{ij}\} \right]^{M_{ij}} \times \prod_{k=1}^{K} \lambda_k^{a-1} \exp(-b\lambda_k) \times \psi_{n,K} \prod_{k=1}^{K} (1 - \sigma)_{m_k - 1}. \tag{8}$$

This expression forms the basis for deriving full conditional updates. In summary, Figure 2 provides a joint representation of the augmented BT–SBM model and its generative process.

5. Posterior inference and single-site Gibbs sampling. Having derived the full posterior distribution in Section 4.5, we now turn to inference. In particular, we employ a single-site Gibbs sampler that exploits the conditional conjugacy of the augmented model. This leads to closed-form full conditional distributions for all variables in Eq. (8), which we detail below.

Auxiliary–variable update. For every observed dyad,

$$Z_{ij} \mid \text{rest} \sim \Gamma\big(n_{ij}, \lambda_{x_i} + \lambda_{x_j}\big). \tag{9}$$

Block–strength update. Let $I_k = \{i : x_i = k\}$ and set

$$w_i = \sum_{j \neq i} w_{ij}, \qquad Z_i = \sum_{j \neq i} Z_{ij}.$$

With a $\Gamma(a, b)$ prior, the full conditional is

$$\lambda_k \mid \text{rest} \sim \Gamma\big(a + \textstyle\sum_{i \in I_k} w_i,\ b + \sum_{i \in I_k} Z_i\big). \tag{10}$$

See Appendix A for further details of this derivation.

Block assignment update. For each item i we sample the label x_i from its unnormalised full conditional

$$p(x_i = k \mid \text{rest}) \propto p(x_i = k \mid \mathbf{x}_{-i}) \times \frac{p\big(\mathbf{W} \mid x_i = k, \mathbf{x}_{-i}, \boldsymbol{\lambda}, \mathbf{Z}\big)}{p\big(\mathbf{W}_{-i} \mid \mathbf{x}_{-i}, \boldsymbol{\lambda}, \mathbf{Z}\big)}, \tag{11}$$

where the prior term follows the urn scheme shown in Section 4.2. In the second term, we have the ratio, whose denominator cancels contributions from pairs not involving i.

For convenience, we denote by $L_i(k)$ the likelihood factor associated with assigning item i to cluster k, i.e. the term multiplying the prior in (11), defined as:

$$L_i(k) \coloneqq \frac{p\big(\mathbf{W} \mid x_i = k, \mathbf{x}_{-i}, \boldsymbol{\lambda}, \mathbf{Z}\big)}{p\big(\mathbf{W}_{-i} \mid \mathbf{x}_{-i}, \boldsymbol{\lambda}, \mathbf{Z}\big)} = \lambda_k^{w_i} e^{-\lambda_k Z_i}, \text{for } k = 1, \ldots, K,$$

with $w_i = \sum_{j \neq i} w_{ij}$ and $Z_i = \sum_{j \neq i} Z_{ij}$. (See Appendix A for details.)

For a new cluster $(k = K + 1)$, we follow Neal (2000)'s Algorithm 3 by integrating out the yet–to–be-sampled λ_{K+1} to obtain:

$$\begin{aligned} L_i(k) &= \int_0^\infty \lambda_k^{w_i} e^{-\lambda_k Z_i} \frac{b^a}{\Gamma(a)} \lambda_k^{a-1} e^{-b\lambda_k} \, d\lambda_k \\ &= \frac{b^a \, \Gamma(a + w_i)}{\Gamma(a)} (b + Z_i)^{-(a + w_i)} \quad \text{for } k = K + 1. \end{aligned} \tag{12}$$

Combining prior and likelihood terms gives

$$p(x_i = k \mid \text{rest}) \propto \begin{cases} (m_{-i,k} + 1)((n-1) - K + \gamma) \, L_i(k), & k = 1, \ldots, K; \\ (K^2 - K\gamma) \, L_i(k), & k = K + 1. \end{cases} \tag{13}$$

Normalising over $k \in \{1, \ldots, K + 1\}$ yields the sampling probabilities. If the draw selects the new label $K+1$, we set $x_i \leftarrow K + 1$, and then we sample

$$\lambda_{K+1} \sim \Gamma\big(a + w_i, b + Z_i\big),$$

and increment the cluster count $K \leftarrow K + 1$. Conversely, K decreases naturally whenever a cluster becomes empty. Note that the number of *occupied* clusters K is a random quantity, determined by the current configuration of $\mathbf{x}$. Unlike reversible-jump or split–merge samplers, this approach explores a variable-dimensional posterior space without explicit trans-dimensional proposals: clusters are created or removed naturally as labels are updated.

5.1. *Hyperparameter specification and scale alignment.* The model has three hyperparameters: (a, b) for the $\Gamma(a, b)$ prior on block strengths and γ for the Gnedin prior on $\mathbf{x}$ (see Section 4.2). Since the Bradley–Terry likelihood depends only on ratios,

$$\frac{\lambda_{x_i}}{\lambda_{x_i} + \lambda_{x_j}} = \frac{c \, \lambda_{x_i}}{c \, \lambda_{x_i} + c \, \lambda_{x_j}}, \quad (i, j) \in E.$$

the overall scale of $\boldsymbol{\lambda}$ is not identifiable. We therefore normalise each iteration by imposing

$$\frac{1}{K}\sum_{k=1}^{K}\log\lambda_k = 0.$$

We choose (a, b) on the same scale. For $\lambda_k \sim \Gamma(a, b)$, setting

$$b = \exp\{\psi(a)\},$$

with ψ the digamma function, makes $\mathbb{E}[\log\lambda_k] = 0$. Once b is fixed in this way, a controls the prior heterogeneity through

$$\mathrm{Var}[\log\lambda_k \mid a, b] = \psi_1(a),$$

where ψ_1 is the trigamma function. In Appendix B we show that lower values of a increase the probability of creating new clusters, while higher values of a have the opposite effect. At the same time, smaller a implies higher variability of $\boldsymbol{\lambda}$, since $\psi_1(a) \approx 1/a$ is strictly decreasing in a, thus yielding a less informative prior. These two forces act in opposite directions, so there is no universally optimal choice of a. We recommend setting $a \in [2, 4]$ and $\gamma = 0.8$, which implies $\mathbb{E}[K] \approx 2.4$ and $\mathrm{Var}[K] \approx 46$ (see Section 4.2). Further motivation and sensitivity analysis are deferred to Appendix B.

To conclude, Algorithm 1 collects the updates described above into a conjugate single-site Gibbs sampler with augmentation for posterior inference under the BT–SBM. A full sweep requires $\mathcal{O}(|E|)$ operations to update the latent variables Z_{ij}, $\mathcal{O}(nK)$ operations for the cluster allocations, and $\mathcal{O}(K)$ operations for the block intensities λ_k. The overall cost per iteration is therefore $\mathcal{O}(|E| + K(n+1))$, which is effectively linear in $|E|$, since the number of occupied blocks typically satisfies $K \ll n$.

We next discuss how to summarise the resulting posterior samples and quantify uncertainty. We then assess the empirical behaviour of the BT–SBM in a simulation study, before turning to the real data application.

Algorithm 1 Conjugate single-site Gibbs sampler with augmentation for the BT–SBM

Require: Initialise at $t = 0$: $x_i^{(0)} = i$ for $i = 1, \ldots, n$, $K^{(0)} = n$, and $\lambda_k^{(0)} \sim \Gamma(a, b)$ independently.

1: **for** $t = 0, \ldots, T-1$ **do**

2: **Update auxiliary variables:** For each (i, j) with $n_{ij} > 0$, sample $Z_{ij}^{(t+1)}$ from (9).

3: **Update strengths:** For each occupied cluster k, sample $\lambda_k^{(t+1)}$ from (10).

4: **Update allocations:** For each $i = 1, \ldots, n$:

1. Remove i from its current cluster; delete empty clusters and relabel if needed.
2. Sample $x_i^{(t+1)}$ using (13) over $k = 1, \ldots, K^{(t)} + 1$.
3. If $x_i^{(t+1)} = K^{(t)} + 1$, create a new cluster, draw its strength from the corresponding Gamma full conditional in (10), and set $K^{(t+1)} \leftarrow K^{(t)} + 1$.

5: **Global rescaling:** Impose $\frac{1}{K^{(t+1)}}\sum_{k=1}^{K^{(t+1)}}\log\lambda_k^{(t+1)} = 0$.

6: **end for**

7: **return** Posterior draws $\{(\mathbf{x}^{(t)}, \boldsymbol{\lambda}^{(t)}, K^{(t)})\}_{t=B+1}^{T}$.

6. Point estimates and uncertainty quantification. Our MCMC sampler yields posterior samples of $\mathbf{x}$ and $\boldsymbol{\lambda}$, which we use to approximate the posterior distribution and derive summary statistics. After collecting the samples $\{\mathbf{x}^{(t)}, \boldsymbol{\lambda}^{(t)}, K^{(t)}\}_{t=1}^{T}$, we discard the first B iterations as burn-in to eliminate the non-stationary phase of the chain. Again, we emphasise that $K^{(t)}$ is the number of occupied clusters in $\mathbf{x}^{(t)}$ and hence it can be easily obtained from the latter.

6.1. *Identifiability.* The model likelihood is invariant under permutations of cluster labels. That is, for any permutation π of the labels,

$$\mathcal{L}(\mathbf{W} \mid \mathbf{x}, \boldsymbol{\lambda}, \mathbf{N}) = \mathcal{L}(\mathbf{W} \mid \mathbf{x}_\pi, \boldsymbol{\lambda}_\pi, \mathbf{N}),$$

where $\mathbf{x}_\pi$ and $\boldsymbol{\lambda}_\pi$ denote the permuted assignment vector and cluster strengths, respectively. This phenomenon is known as the *label switching problem*, and it makes direct posterior summaries of $\mathbf{x}$ or $\boldsymbol{\lambda}$ meaningless unless the samples are relabelled consistently.

To address this, we relabel each sample by sorting the occupied blocks in decreasing order of strength: the block with the largest $\lambda_k^{(t)}$ receives label 1, the second largest label 2, and so on. This post-processing aligns all samples to a common convention; a short pseudocode summary is reported in Appendix B.

The BT–SBM likelihood is also invariant under multiplicative rescaling of the block strengths, and this issue is addressed above (see Section 5.1).

6.2. *Posterior summaries, uncertainty quantification and model choice.* Unlike algorithmic clustering methods that return a single point estimate, the BT–SBM yields a full posterior distribution that allows for uncertainty quantification. We summarise this posterior across the latent partition $\mathbf{x}$, the number of clusters K, and the strength parameters $\boldsymbol{\lambda}$.

6.2.1. *Partition estimation.* To obtain a representative clustering, we adopt a decision-theoretic approach based on the Variation of Information (VI) metric (Meilă, 2007; Wade and Ghahramani, 2018). The variation of information (VI) quantifies the discrepancy between two partitions through their marginal and joint entropies. It can be interpreted as the amount of information lost when replacing one clustering with another, and it attains the value zero if and only if the two partitions coincide. We define the point estimate $\widehat{\mathbf{x}}$ as the partition that minimises the posterior expected VI distance from the sampled clusterings,

$$\widehat{\mathbf{x}} = \arg\min_{\mathbf{x}'} \mathbb{E}[\mathrm{VI}(\mathbf{x}, \mathbf{x}') \mid \mathbf{W}],$$

with the expectation approximated using relabelled MCMC draws, via the `mcclust` and `mcclust.ext` implementations (Wade and Ghahramani, 2018). Empirically, VI-based summaries favour parsimonious but stable partitions.

To quantify uncertainty, we construct a $(1-\alpha)$ credible ball under the VI metric, defined as the smallest region around $\widehat{\mathbf{x}}$ containing $1-\alpha$ posterior mass (we use $\alpha = 0.05$). Because the space of partitions is discrete and high-dimensional, we summarise this region through its representative *surface* partitions (vertical upper, vertical lower, and horizontal bounds), which characterise the coarsest, finest, and intermediate resolutions supported by the posterior. Full computational details can be found in Wade and Ghahramani (2018), and are also reported in Appendix E for the reader's convenience.

6.2.2. *Number of clusters.* Uncertainty on K is summarised directly from the empirical distribution of $\{K^{(t)}\}$. We report the posterior mode $\widehat{K}$ together with a 95% credible interval. In addition, we consider the number of clusters implied by $\widehat{\mathbf{x}}$ and by the VI credible ball. In practice, these quantities do not need to agree, as they are substantially different in nature.

6.2.3. *Strength parameters.* Block-level strengths are summarised by posterior means computed from relabelled draws. Player-level strengths are obtained by mapping each player to the strength of its assigned block at each iteration, that is, $\tilde{\lambda}_i^{(t)} := \lambda_{x_i}^{(t)}$. The associated vector is $\tilde{\boldsymbol{\lambda}} := (\tilde{\lambda}_1, \ldots, \tilde{\lambda}_n)$. Credible intervals are obtained directly from the corresponding posterior samples. Together, these summaries and their associated credible sets allow us to represent both the underlying hierarchical structure of the network and the corresponding uncertainty within a coherent Bayesian framework.

7. Simulation study: main results. In this section, we highlight the most important results from our simulation study (which is reported in full detail in Appendix D). We assess whether the BT–SBM can recover the latent partition $\mathbf{x}$, strengths $\boldsymbol{\lambda}$, and number of clusters K, under a challenging scenario that closely resembles the ATP data studied in the present paper. We generate synthetic win-loss pairwise data $\mathbf{W}$ starting from the match-count matrix $\mathbf{N}^{(s)} = (n_{ij})$ from three representative seasons $s \in \{\text{“2000”}, \text{“2015”}, \text{“2024”}\}$, so that each synthetic dataset inherits the structural features discussed in Section 2, such as the pronounced and uneven sparsity pattern.

For each s, we generate datasets with $K^\star \in \{3, 5, 7\}$, for a total of 9 datasets with different $(s \times K^\star)$ combinations. Partitions $\mathbf{x}^\star$ are rejection-sampled from a Gnedin prior, conditional on $K(\mathbf{x}^\star) = K^\star$ (Appendix D). Block strengths are set to yield adjacent-tier separation $p(k \text{ beats } k{+}1) \approx 0.8$ under the BT likelihood, and are normalised to satisfy $\frac{1}{K^\star}\sum_{k=1}^{K^\star} \log \lambda_k = 0$. Conditional on $(N^{(s)}, \mathbf{x}^\star, \boldsymbol{\lambda}^\star)$, $\mathbf{W}$ is sampled from the Binomial BT–SBM likelihood (6). Full generative details needed to reproduce datasets with features similar to those we observe in the ATP dataset (like the winners-play-more effect) are discussed in Appendix D.

7.1. *Recovery performance.* We fit the correctly specified BT–SBM to the nine synthetic datasets using $T = 5000$ Gibbs iterations with burn-in $B = 2000$ (fixed across scenarios). Recovery is assessed via relative mean absolute error ($\mathrm{MAE}_{\mathrm{rel}}$) on player-level strengths, Spearman rank correlation between true and estimated strengths ($\rho(\widehat{\tilde{\boldsymbol{\lambda}}}, \tilde{\boldsymbol{\lambda}}^\star)$), and partition agreement between $\widehat{\mathbf{x}}$ and $\mathbf{x}^\star$ measured by ARI (Hubert and Arabie, 1985) and VI (Meilă, 2007) (definitions and further diagnostics are deferred to Appendix D).

Table 1 reports averages across the three scenarios considered above. Recovery is strong for $K^\star = 3$ and remains accurate for $K^\star = 5$. To provide more detail for $K^\star = 3$, the true block strengths $(0.135,\ 1.000,\ 2.356)$ are closely matched by their corresponding posterior means $(0.149,\ 0.972,\ 2.349)$ with posterior standard deviations $(0.019,\ 0.067,\ 0.137)$, respectively.

TABLE 1
BT–SBM recovery by target number of clusters $K^\star$, averaged across exposure designs $s \in \{\text{“2000”}, \text{“2015”}, \text{“2024”}\}$.

$K^\star$	$\mathrm{MAE}_{\mathrm{rel}}$	$\rho(\widehat{\tilde{\boldsymbol{\lambda}}}, \tilde{\boldsymbol{\lambda}}^\star)$	$\mathrm{ARI}(\widehat{\mathbf{x}}, \mathbf{x}^\star)$	$\mathrm{VI}(\widehat{\mathbf{x}}, \mathbf{x}^\star)$	$\widehat{K}$ (95% CI)
3	0.32	0.84	0.90	0.29	$3_{[3,5]}$
5	0.41	0.92	0.80	0.78	$5_{[4,7]}$
7	0.44	0.92	0.60	1.11	$6_{[4,8]}$

For $K^\star = 7$, the global ordering is still well identified (high $\rho(\widehat{\tilde{\boldsymbol{\lambda}}}, \tilde{\boldsymbol{\lambda}}^\star)$), although the posterior cannot always distinguish between adjacent blocks. In addition, the average number of edges per block decreases with $K^\star$ when $(n, |E|)$ is fixed, resulting in skewed cluster sizes and edge distributions.

7.2. *Comparison with alternative models.* We compare BT–SBM to the classical BT model as well as to the rank-clustering Bradley–Terry model (RCBTL) of Pearce and Erosheva (2025). BT, RCBTL and BT–SBM are run for $T = 15{,}000$ iterations with burn-in $B = 5{,}000$, matching the number of iterations in Pearce and Erosheva (2025). RCBTL is specified so that $\mathbb{E}[K] = 2$ by setting $\lambda^{\mathrm{RCBTL}} = 2$, that is, the Poisson hyperparameter on K_g in their model as mentioned in Section 3. Similarly, BT–SBM has $\mathbb{E}[K] \approx 2.4$ by setting $\gamma = 0.8$ in the Gnedin prior. Convergence diagnostics are computed by running four independent chains, as recommended by Pearce and Erosheva (2025). More details on initialisations and hyperparameters can be found in Appendix D.

We report an illustrative comparison on a single dataset ($s =$ "2000", $K^\star = 3$). Table 2 summarises the inferred number of blocks, partition and strength recovery (when applicable). BT–SBM yields the strongest partition recovery and the inferred number of clusters concentrates near the true $K^\star$. RCBTL shows a comparable strength correlation but substantially overestimates the number of clusters in this instance, ARI and VI signal that it achieves a worse partition recovery, and is markedly slower (roughly 1560 times slower). Full diagnostics (including Gelman–Rubin convergence diagnostics) are reported in Appendix D.

TABLE 2
Baseline comparison on one synthetic dataset ($s =$ "2000", $K^\star = 3$, $n = 105$).

Model	$\widehat{K}$ (95% CI)	$\rho(\widehat{\tilde{\boldsymbol{\lambda}}}, \tilde{\boldsymbol{\lambda}}^\star)$	ARI$(\widehat{\mathbf{x}}, \mathbf{x}^\star)$	VI$(\widehat{\mathbf{x}}, \mathbf{x}^\star)$	Time
BT	–	0.863	–	–	44.4 s
BT–SBM	3 $[3, 5]$	0.874	0.964	0.134	50.3 s
RCBTL	17 $[14, 20]$	0.873	0.820	0.765	77,980 s

8. Analysis of the last 26 years of men's tennis. We now present a detailed analysis of the 2017 ATP men's tennis season, highlighting the main outputs of our model. We focus on this season because of the substantial uncertainty regarding the number of clusters. This case study also serves to illustrate the BT–SBM more concretely in the context of real competition data.

We subsequently repeat the analysis across all seasons from 2000 to 2025 (26 seasons in total), a period spanning the rise, peak, and eventual decline of the dominance exerted by Federer, Nadal, Djokovic, and Murray—the quartet commonly referred to as the "Big Four."

This era profoundly shaped men's tennis, and our model captures its evolution through changes in the inferred competitive structure over time. For each season, our MCMC algorithm (described in Section 5) was independently run for 30,000 iterations, discarding the first 10,000 as burn-in. Fitting the model to all 26 seasons under these specifications required approximately 35 minutes in total. We used a Gnedin prior for $\mathbf{x}$ with hyperparameter $\gamma = 0.8$, which implies $\mathbb{E}[K] \approx 2.4, \mathrm{Var}[K] \approx 46$, and a Gamma prior for the rate parameters with shape $a = 2$ and rate $b = \exp\{\psi(a)\} = 1.526$.

Following this analysis, we relabelled posterior draws by ordering occupied blocks from strongest to weakest. All computations, for both the simulation study and the real-data application, were implemented in `R`; code and data are available through the repository described in Section 9.

8.1. *The 2017 ATP men's season.* We examine in detail the 2017 ATP season. Table 3 reports posterior probabilities for the number of blocks under the Gnedin–type prior. The posterior slightly favours a four-block model ($\widehat{K} = 4$), although the 95% credible interval spans configurations between 3 and 7 clusters.

TABLE 3
Posterior distribution of the number of clusters K for the 2017 ATP season under the Gnedin–type prior. The modal configuration is $K = 4$, but substantial mass over $K \in \{3, \ldots, 7\}$ indicates non-negligible uncertainty in the latent resolution.

Season	2	3	4	5	6	7	8
2017	–	0.259	**0.315**	0.216	0.110	0.056	0.025

To obtain a representative partition, we compute the VI-loss point estimate $\widehat{\mathbf{x}}$ (Section 6.2) and reorder the adjacency matrix accordingly (Fig. 3). As VI penalises small clusters, it yields

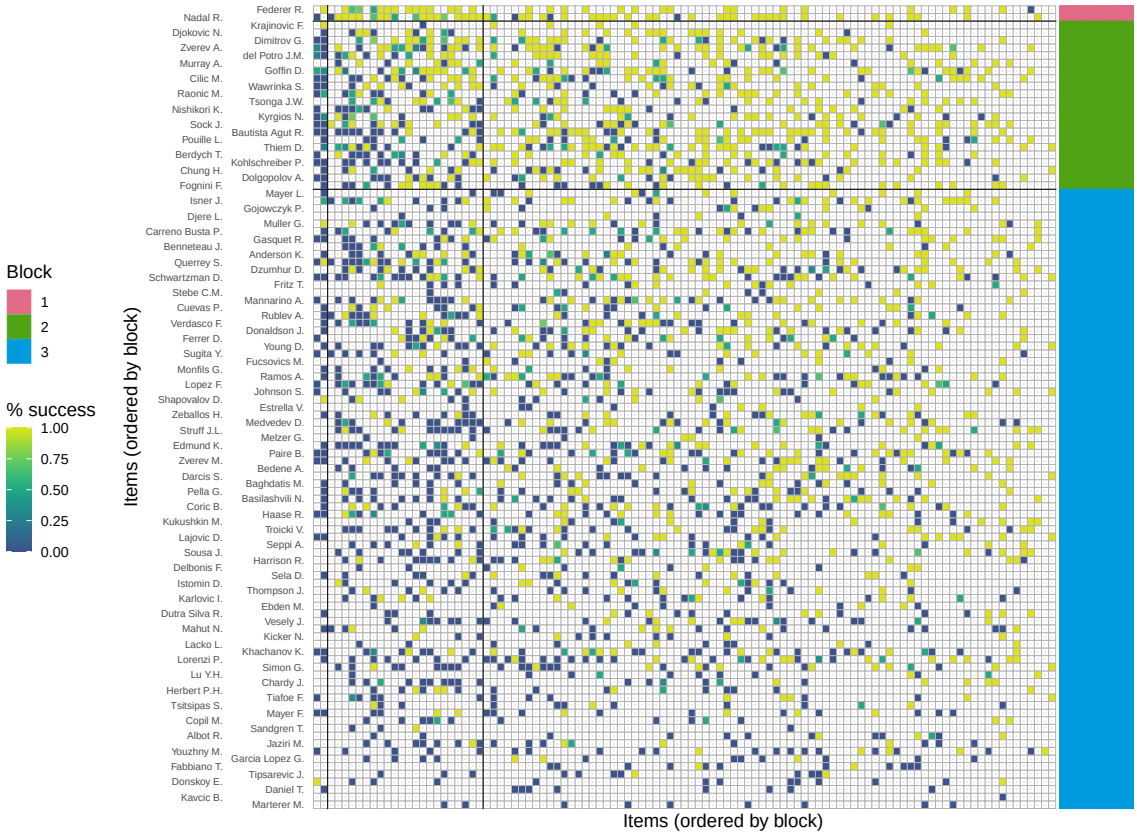


Fig 3: Reordered adjacency matrix of ATP matches during the 2017 season, obtained from $\widehat{\mathbf{x}}$ with $K = 3$. Players are ordered by inferred cluster (1 = strongest to 3 = weakest) and then by marginal win proportion. Each pixel shows the pairwise win rate (W_{ij}/n_{ij}). Mixed colours along the diagonal indicate within-block balance, while the dark top-right (and light bottom-left) regions reflect systematic dominance of stronger tiers. Federer corresponds to the first row, Nadal to the second, Krajinovic to the third, and so on.

a parsimonious configuration with $\widehat{K}^{(\mathrm{VI})} = 3 < \widehat{K} = 4$. Appendix E reports an alternative representative $K = 4$ partition obtained from a tuned generalised VI loss. The induced partition separates the players into three tiers of markedly different competitive strength.

Block 1 isolates Federer and Nadal, reflecting the pronounced competitive gap of 2017. This dominance is visible in the top-right corner of the reordered adjacency matrix, where dark-green pixels correspond to head-to-head winning probabilities close to one. Posterior estimates of the model-implied inter-block win probabilities quantify this separation: players in block 1 defeat block 2 with estimated probability approximately equal to 0.79 and block 3 with estimated probability approximately equal to 0.91, while block 2 defeats block 3 with estimated probability approximately equal to 0.72.

Along the main diagonal of Fig. 3, we observe matches between players belonging to the same block. These regions exhibit win probabilities close to 0.5, as indicated by the mixed light and dark colours. The resulting structure is therefore sharply hierarchical across blocks and approximately homogeneous within blocks.

Although the VI partition fixes $K = 3$, posterior uncertainty on K remains substantial. Appendix E reports that the 95% VI credible ball still contains finer partitions, with support extending up to $K = 8$. What changes is mostly the resolution, not the order: the elite block stays stable, while extra blocks appear at the boundaries of the middle and lower tiers. The

vertical and horizontal lower bound partitions and their reordered adjacency matrices are also reported in Appendix E.

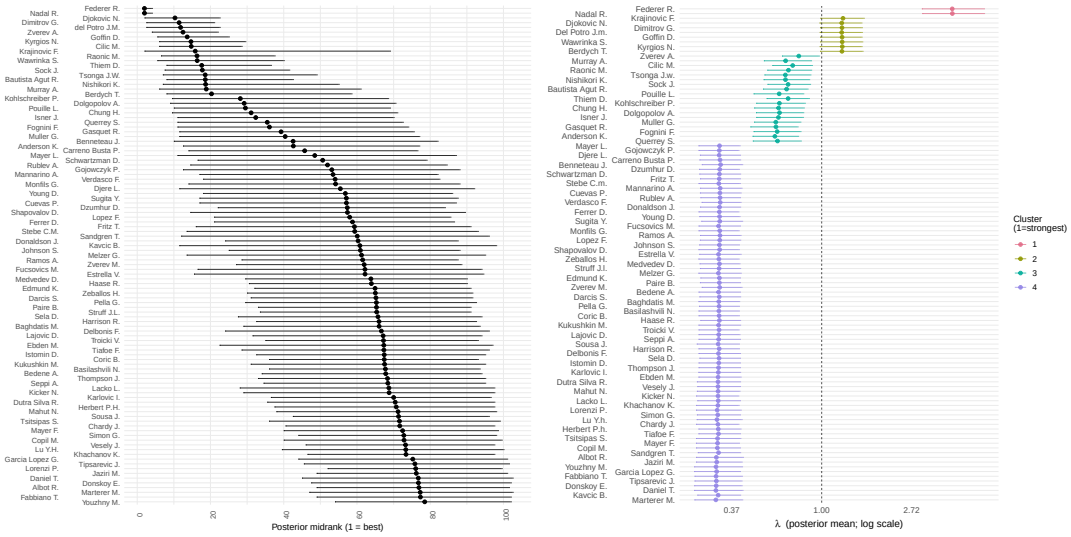
In the limit, the classical Bradley–Terry model corresponds to $K = n$, enforcing full separation irrespective of evidential support. The BT–SBM instead regularises this tendency by allowing players to share latent strength parameters, recovering a stable tiered structure while propagating uncertainty in the number and location of block boundaries. For the 2017 season, the evidence consistently supports a small elite, a distinct middle tier, and a broad lower tier, separated by large and statistically meaningful inter-block win probabilities.

Uncertainty in cluster assignments. Figure 4 shows posterior membership probabilities $p(x_i = k \mid \mathbf{W})$, restricted to MCMC draws with $K^{(t)} = 4$, thus isolating allocation uncertainty from uncertainty on K. Uncertainty is driven by (i) limited data, affecting players at the bottom of the hierarchy, and (ii) discontinuous performance among highly observed players. The top tier shows near-deterministic assignments, while uncertainty concentrates between blocks 2 and 3 and at the bottom of block 3, where several players exhibit diffuse memberships compatible with an additional group.



Fig 4: Posterior probabilities $p(x_i = k \mid \mathbf{W})$ for the 2017 ATP season, conditioned on draws with $K = 4$. Players are ordered as in Fig. 3. Darker shades indicate higher posterior mass. Uncertainty concentrates between blocks 2 and 3 and at the bottom of block 3, where several players have diffuse membership compatible with a fourth block.

Player-level strength uncertainty. Here we illustrate how the inferred posterior strengths of each player across all iterations can be used to assess an implied posterior ranking of players. Since players within the same block share a common strength parameter, the induced ordering at each MCMC iteration yields a *tied* player ranking. Let $\tilde{\lambda}_i^{(t)}$ denote the strength associated with player i at iteration t, that is, the strength of the block i is currently assigned to. Sorting



(a) Posterior mean midranks marginalised over K (b) Posterior mean strengths conditional on $K = 4$. (1 = best).

Fig 5: Player ranking summaries estimated marginally over K and conditionally on $K = 4$. The marginal panel reports posterior midranks, while the conditional panel fixes K and emphasises separation between tiers on the λ scale.

$\{\tilde{\lambda}_i^{(t)}\}_{i=1}^n$ in decreasing order produces a rank ordered list of players. However, whenever several players belong to the same block they share the same value of $\tilde{\lambda}_i^{(t)}$ and are therefore tied.

To summarise such tied rankings we transform the tied ranks into *midranks*, the tied players' average rank. At iteration t, suppose a tied group occupies integer positions $a, \ldots, b$ in the sorted list, with size $b - a + 1$. Each player in the group is then assigned $r_i^{(t)} = (a + b)/2$. For example, a ranking of the form $\{1, 2, 2, 3, 3, 3\}$ becomes $\{1, 2.5, 2.5, 5, 5, 5\}$ under the midrank transformation.

Midranks are convenient because they treat tied players symmetrically, avoiding any arbitrary within-block ordering. Moreover, they also preserve the total rank mass at every iteration, $\sum_{i=1}^n r_i^{(t)} = (n(n+1))/2$ for all t, so that such transformed ranks live on a scale independent of K. Collecting $\{r_i^{(t)}\}_{t=B+1}^T$ therefore yields a posterior distribution of midranks for each player.

Figure 5a displays the marginal midrank estimator $\widehat{r}_i$, obtained by averaging over uncertainty on $\mathbf{x}$, K, and $\boldsymbol{\lambda}$. The presence of ties at the player level and relatively wide credible intervals reflect the structural feature of the data discussed in Section 2. When head-to-head outcomes are characterised by recurrent upsets, players follow heterogeneous tournament schedules, participation in major tournaments is imbalanced, and unpredictable events such as injuries or luck intervene, the resulting evidence does not justify fine-grained rank distinctions.

In this perspective, a maximum posterior mean midrank around 80 is entirely consistent with $n = 105$ players and with the existence of a broad bottom tier, the one visible in Fig. 3. In short, the dispersion observed in Fig. 5a is a quantitative expression of the same phenomenon discussed in the introduction, namely that fine-grained rank distinctions are often more precise than the data can credibly sustain.

Figure 5b reports the conditional estimator $\widehat{\tilde{\boldsymbol{\lambda}}}$ computed using only those iterations with $K^{(t)} = 4$, thereby isolating the variability in $\boldsymbol{\lambda}$ from that in K. The plot shows limited

heterogeneity among players assigned to the same block, indicating that much of the dispersion observed in the marginal summary is driven by iterations with $K \neq 4$. The conditional representation therefore acts both as a diagnostic device and as a clearer visualisation of cluster separation at a fixed resolution.

To conclude, in both representations Nadal and Federer form a distinct upper cluster, clearly separated from the remainder of the field. The middle region exhibits overlapping credible intervals and moderate dispersion, consistent with the partition-level uncertainty discussed above and detailed in Appendix E. The lower tier displays comparatively compressed strengths, reflecting both competitive similarity and limited information to further discriminate among players.

8.2. *BT–SBM analysis: all seasons* 2000–2025. We now apply our model independently to each season of the first quarter of the 21st century, a period spanning the rise, dominance, and gradual decline of the so-called *Big Four*.[2]

We find that early seasons place most posterior mass on $K = 3$; during the Big Four years that mass shifts towards $K = 4$; in 2019, the modal solution moves back mostly to $K = 3$, but with more diffuse probabilities and a broader elite before tightening again in 2023–2025, when Alcaraz and Sinner rise to the top of the world rankings.

Appendix F reports the full season-by-season posterior probabilities for K. These probabilities are informative beyond the modal value of K: in some seasons they are tightly concentrated, whereas in others they spread substantial mass across adjacent values – at times extending past $K = 5$ – signalling uncertainty about how finely the competitive hierarchy should be resolved.

The same appendix also reports a season-by-season predictive comparison against the classical Bradley–Terry model. The BT–SBM model outperforms standard BT in every season we study, with a median gain of about 22.7 ELPD points (Vehtari, Gelman and Gabry, 2017), which supports the block representation empirically. Our main substantive question, however, concerns competitive balance over time. In our setting, competitive balance refers to how concentrated the top tier is and how sharply it is separated from the rest of the field. We therefore study how this balance changes from one season to the next through the posterior probability that a player belongs to the top block.

Measuring the evolution of competitive balance. The posterior probability that a player belongs to the top block measures how certain is the allocation of a player to the strongest tier. When these probabilities concentrate near 0 or 1, the elite is sharply delineated: players are either clearly "in" or clearly "out". When they are more dispersed, several players occupy an intermediate position, which points to a broader and more contested elite. We begin by summarising this uncertainty through the Shannon entropy associated with top-block membership at each MCMC iteration t.

Specifically, at iteration t we compute the *model-based* probability for player i to belong to the top block,

$$q_i^{(t)} := p(x_i^{(t)} = 1 \mid \text{rest}),$$

obtained by computing the full conditional in (13) on the relabelled draws. We then summarise the overall uncertainty in top-block membership at iteration t through the average binary entropy

$$H^{(t)} = -\frac{1}{n}\sum_{i=1}^{n}\Big(q_i^{(t)}\log q_i^{(t)} + (1-q_i^{(t)})\log(1-q_i^{(t)})\Big). \tag{14}$$

[2]With this term, the media refer to Federer, Nadal, Djokovic, and Murray, who captured the lion's share of major titles throughout that period, thereby profoundly shaping the competitive structure of the men's tour.

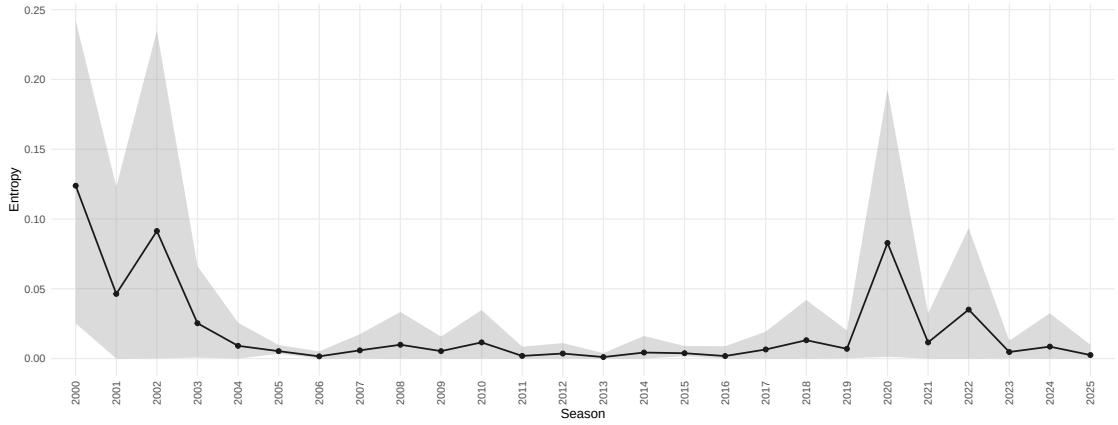


Fig 6: Posterior mean Shannon entropy of top-cluster composition across seasons, with 95% credible intervals. High values indicate a broader and more contested elite, whereas low values reflect concentrated dominance.

This entropy is maximised when many players have $q_i^{(t)} \approx 1/2$, and approaches 0 when most probabilities are near 0 or 1. In our setting, low entropy indicates a sharp separation around the top block (players are either clearly "in" or clearly "out"), consistent with a more concentrated and stable top tier, whereas high entropy reflects greater uncertainty and turnover in top-block membership. For interpretability across seasons, we report the normalised entropy

$$\widetilde{H}^{(t)} = \frac{H^{(t)}}{\log(n)} \in [0, 1],$$

so that values near 0 correspond to a sharply defined elite, with essentially no uncertainty about who belongs to it. Values closer to 1 instead indicate a more porous and contested top tier, in which competition for those places is still unresolved.

Figure 6 displays the posterior mean normalised entropy over time, with 95% credible intervals. Read together with the posterior distribution of K in Appendix F, and the size of the top block in Fig. 7, it reveals a clear temporal pattern that can be summarised into four broad phases.

Early 2000s. Between 2000 and 2005, the three-block solution is consistently dominant, suggesting a relatively fluid and competitive upper tier. Figure 7 supports this reading: the top cluster is noticeably larger, multiple players rotate through it, and the relatively high entropy values in Figure 6 reinforce the idea of a contested elite.

Mid 2000s to 2018. From 2006 onward, the model increasingly favours four-block solutions, and the top cluster shrinks abruptly, marking the emergence and consolidation of the Big Four – a tightly held elite with little rotation. This is visible both in the reduced top-cluster size (Fig. 7) and in the low entropy values (Fig. 6).

From 2019 to 2022. In 2019, the pattern loosens again: the elite broadens, and the rise in entropy in Figure 6 indicates renewed uncertainty in top-block membership. This is a transition phase, shaped by the decline of the Big Four and by the disruption around the pandemic, and it peaks in 2022.

From 2023 to 2025. The analysis of these last three seasons shows a marked shrinking of the elite group. Although the three-block solution remains preferred, non-trivial mass sits on $K = 4$ and $K = 5$, and the two-block solution is never sampled.. Figure 7 shows the renewed contraction of the top block, while Figure 6 shows a pronounced drop in entropy after 2022,

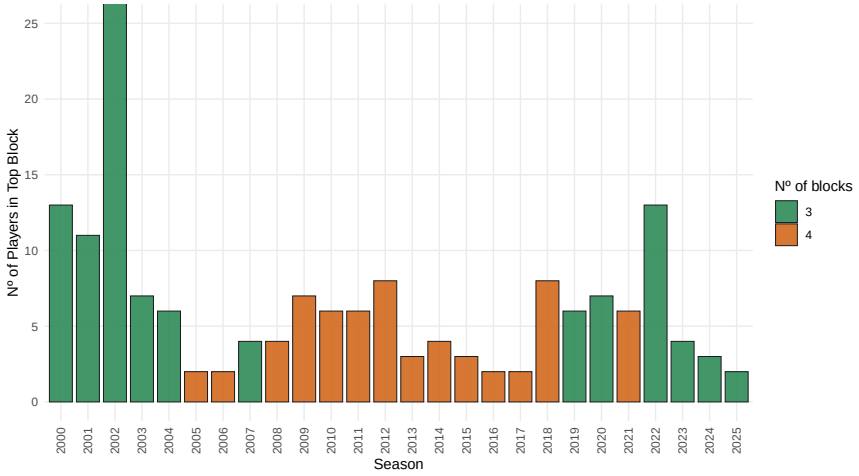


Fig 7: Estimated number of players assigned to the top cluster in each ATP season from 2000 to 2025. The top block is broader in the early 2000s, contracts during the Big Four era, broadens again in 2019–2022, and tightens in 2023–2025.

consistent with the emergence of a new concentrated elite centred on Jannik Sinner and Carlos Alcaraz.

Taken together, the season-by-season posterior summaries in Appendix F, Figure 7, and Figure 6 point to a coherent narrative: the Big Four era is characterised by a tightly concentrated elite and low uncertainty in top-block membership; during the 2019–2022 transition, that structure dissolves and the field reopens; in recent seasons, Carlos Alcaraz and Jannik Sinner have emerged and consolidated as the new elite at the top of the rankings.

9. Conclusion and future directions. We introduced a Bayesian stochastic block model for paired comparisons that sits inside the Bradley–Terry formulation. Players in the same block share the same strength, so the model replaces a strict player-by-player ranking with an ordered set of tiers. This gives an interpretable summary of competitive structure and, at the same time, propagates uncertainty on both strengths and block boundaries.

In men's tennis, this view recovers a clear temporal story. The early 2000s look like a broader elite, the Big Four era like a contracted and sharply separated top tier, and the post-2018 period like a transition that later tightens again around a new elite led by Alcaraz and Sinner. The main limitation is that we condition on observed match counts and keep the Bradley–Terry one-dimensional strength assumption. Natural extensions are to introduce temporal dependence, add covariates such as surface or tournament category, and relax strict transitivity when context-specific effects matter.

Although our application is tennis, the same construction applies to other directed pairwise comparison settings, from animal dominance to preference data and AI evaluation (Baldassarre et al., 2023; Loewen, Rubenson and Spirling, 2012; Hatzinger and Mazanec, 2007; Chiang et al., 2024). The payoff is the same in each case: we recover stable tiers when the data support them, and we keep uncertainty visible when the boundaries between tiers are hard to pin down.

Code and reproducibility. All code to reproduce the analyses, figures, and tables is openly available through the project repository, github.com/laposanti/BT-SBM-Bradley-Terry-Stochastic-Block-Model, and the `BTSBM` package site, laposanti.github.io/BTSBM/.

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Significance statement. Ranking competitors from head-to-head outcomes is central in sports analytics and many paired-comparison problems, but standard approaches often impose a strict complete ordering that overstates small differences, and is not robust to data imbalances. We propose the Bradley–Terry Stochastic Block Model, which clusters competitors into strength tiers while learning the number of tiers from the data. The Bayesian formulation quantifies uncertainty in both tier membership, the number of tiers, and strength levels, and it remains effective under sparse and highly imbalanced schedules typical of tournaments. Applied to 2000–2025 men's professional tennis, the model identifies a persistent elite tier and quantifies changes in competitive balance over time. The approach is broadly applicable to pairwise comparison data beyond sport.

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